Data Analysis and Exploratory Analysis

Analysis of Employee Attrition in an Age of AI

To: Director of Workforce Analytics, Insight Partners
From: Kevin Huang, Junior Data Analyst
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Subject: Analysis into Employee Attrition during an Age of AI

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1. Background and Context

Insight Partners has been working with a mid-to-large enterprise concerned with rising employee attrition in this age of AI. Leadership suspects burnout, stalled promotions, hybrid and remote arrangements, and employees' fear of automation replacing them all contribute, but don't know which matter most or how they interact. Understanding these drivers will help guide employee retention endeavors and workforce policy decisions such as how to support employees and how to structure future work arrangements. Data is audited and cleaned first to prevent misleading decisions before an exploratory analysis to identify early warning indicators for employee attrition.

2. Data Quality Assessment and Preparation

2.1 Data Quality Audit

Table 1 below summarises issues identified in the data however three issues matter the most when analysing employee attrition. Firstly, the 37 duplicated records would double-count employees. Secondly, the 25 missing and 28 invalid values affect a few rows however dropping them completely could skew the results as well. Thirdly, 46 rows have years at company greater than total years worked which would distort any insights drawn in relation to it. A fourth issue is the 673 Remote workers with commute minutes above 0, however this could be attributed to occasional office visits rather than errors, so it was only flagged.

Table 1: Data quality issues identified in the raw dataset

Issue	Variable(s)	Records affected	
		Count	Percentage
Missing values	work_mode	12	0.32%
	stock_option_level	6	0.16%
	monthly_income	3	0.08%
	salary_hike_pct	3	0.08%
	years_at_company	1	0.03%
Duplicates	All variables	37	0.97%
Invalid values	age	17	0.45%
	job_level	11	0.29%
Logistical Inconsistency	years_at_company > total_working_years	46	1.2%
	Remote but commute_minutes > 0	673	87.7%

2.2 Data Cleaning and Preparation

Duplicates were removed as they are whole-row duplicates and their higher attrition rate (21.6% vs 17.9% for unique rows) confirms retention would inflate the attrition rate. The 52 rows with a missing or invalid value have a lower attrition rate (13.5% vs 18.0% for full rows), so dropping them would also exaggerate attrition rate so imputation was used instead. Median imputation was used as income is right-skewed, imputing with mean would introduce

values above the typical employee. The 46 tenure inconsistencies were fixed by raising total working years to the tenure to preserve the attrition signal. After cleaning, the headline attrition rate changed by -0.03 pp and the largest change in any driver correlation was 0.010, so conclusions drawn are not sensitive to cleaning approach.

Table 2: Evidence supporting cleaning decisions (pre-cleaning)

Issue	Decision	Evidence
Duplicates (37 rows)	Remove	• Attrition rate 21.6% vs 17.9% for unique rows
Missing/Invalid Values (52 rows)	Impute	• Attrition rate 13.5% vs 18% for rows with no invalid/missing values. • Monthly income mean ($6998) > median ($6589) -> right-skew
Tenure inconsistency (46 rows)	Raise total working years	• Minimum-change fix which preserves tenure signal
Remote commute (673 rows)	Flag only	• 87.7% Remote workers affected. Variable is unused for analysis

3. Exploratory Data Analysis

3.1 Summary Statistics

The typical employee looks healthy with a mean burnout score of 4.03, however, it masks the problem that overtime hours reach a max of 34.9, indicating potential risk concentration in a minority. Income is right skewed (mean $7,002 vs median $6,593), so averages overstate typical income, and with no performance rating of 1, there is little to say on weakest performers.

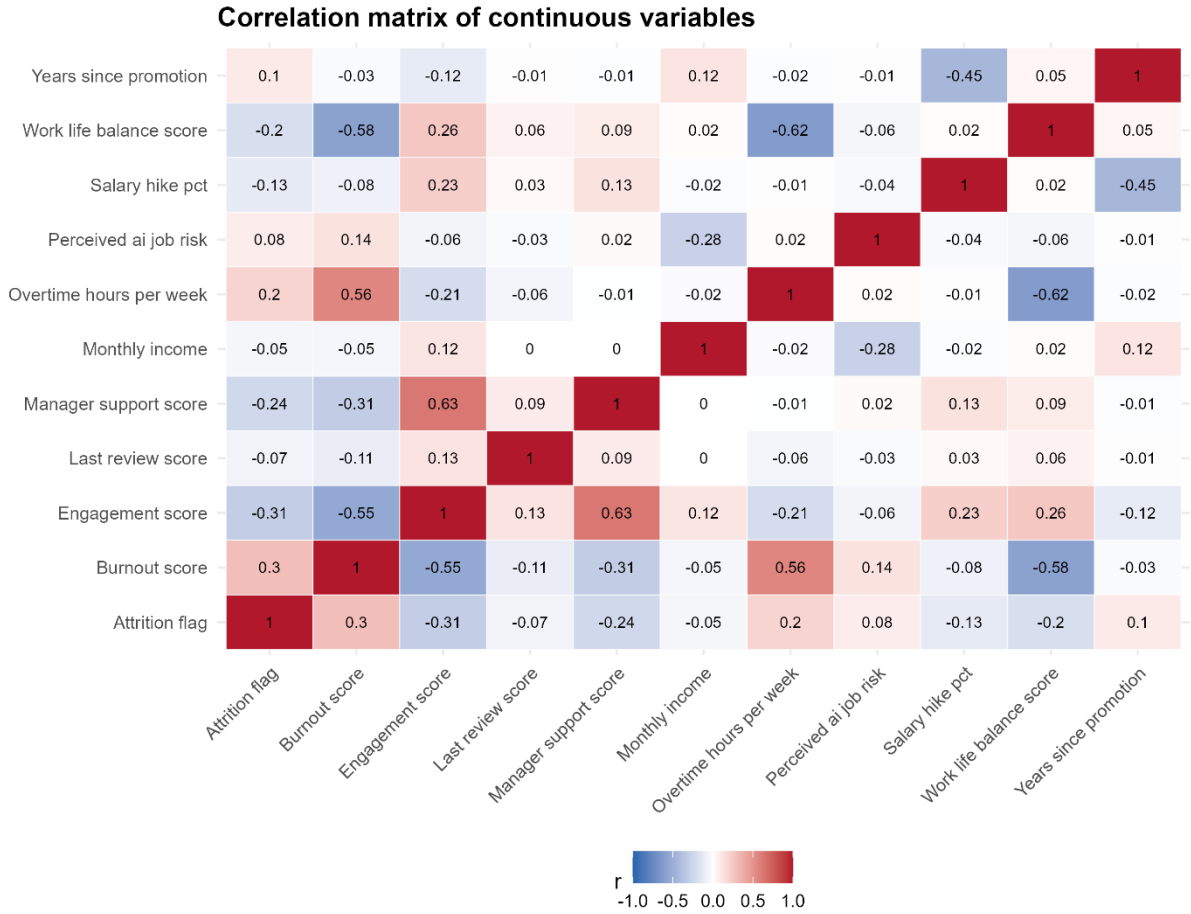
Table 3: Summary Statistics on Key Variables

Variable	Mean	SD	Min	Q1	Median	Q3	Max
age	37.14	8.22	21	31	37	43	64
monthly_income	7001.73	2684.28	1500	4982.5	6593	8643	21609
years_at_company	6.76	3.52	0	4.3	6.8	9.25	18.1
years_since_promotion	1.97	1.62	0	0.8	1.6	2.8	11.1
overtime_hours_per_week	6.82	4.44	0	3.6	5.9	9	34.9
manager_support_score	6.64	1.82	1	5.4	6.7	7.9	10
burnout_score	4.03	1.54	1	2.9	4	5	10
engagement_score	6.54	1.49	1	5.5	6.6	7.6	10
work_life_balance_score	6.12	1.28	1	5.3	6.2	7	9.9
performance_rating	3.63	0.66	2	3	4	4	5
last_review_score	8.33	1.15	2.8	7.5	8.4	9.2	10
perceived_ai_job_risk	4.86	1.82	1	3.6	4.9	6.1	10
attrition_rate_pct	17.86	38.31	0				100

3.2 Correlation Analysis

No single variable explains attrition as all |r| ≤ 0.31, suggesting attrition risk comes from the combination of many factors. Burnout has high correlation to overtime with r = 0.56 whilst decreasing as work-life balance, engagement and manager support scores increase. Furthermore, these variables are interdependent, making it difficult to pinpoint which of them specifically drives attrition. AI risk is weakly linked to attrition with r = 0.08, suggesting an indirect effect.

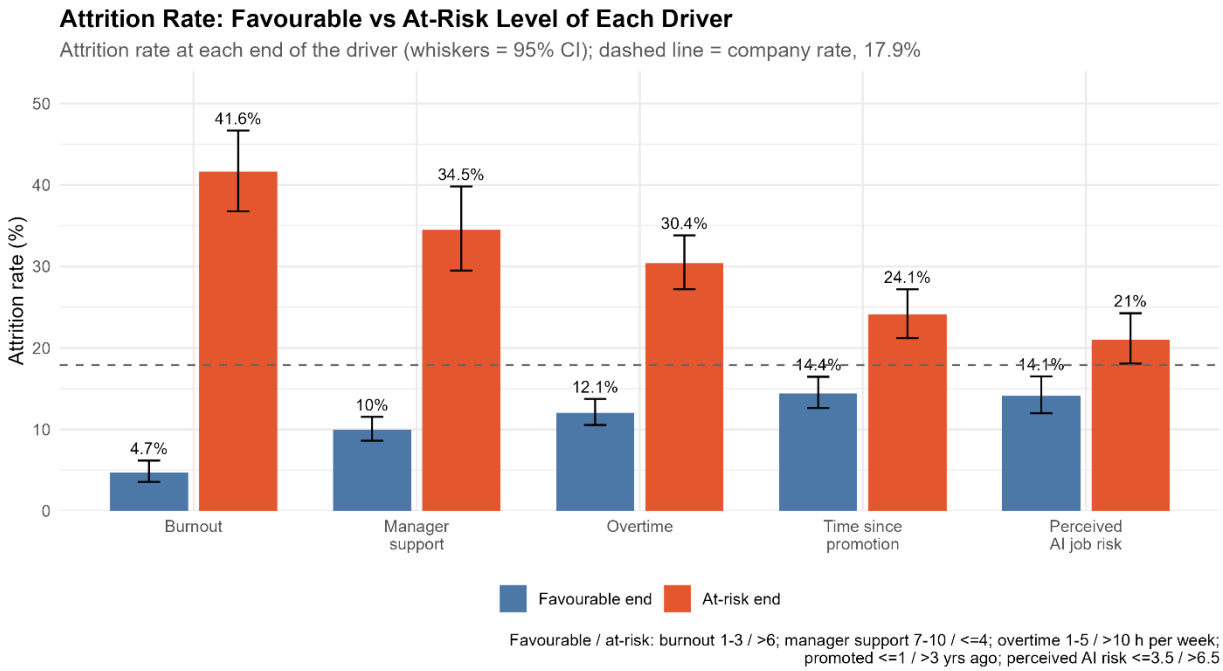
Figure 1: Burnout sits at the center of workplace strain



4. Findings and Insights

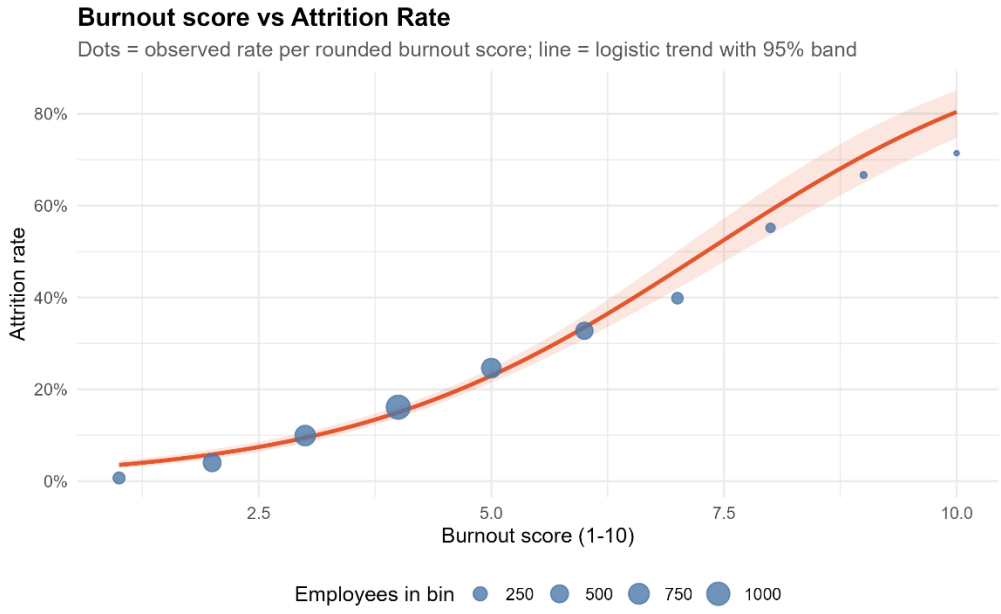
Attrition stems from insufficient managerial support and high burnout rather than by AI risk. Burnout and manager support separate leavers and stayers most with massive attrition rate differences of 36.9% and 24.5% respectively. Both are within management's control to fix. In contrast, AI risk has the smallest difference of only 6.9%. If management focuses investments into AI-reassurance initiatives, they would target the smallest driver while leaving the major underlying workforce strains unattended.

Figure 2: Burnout and manager support separate leavers from stayers the most



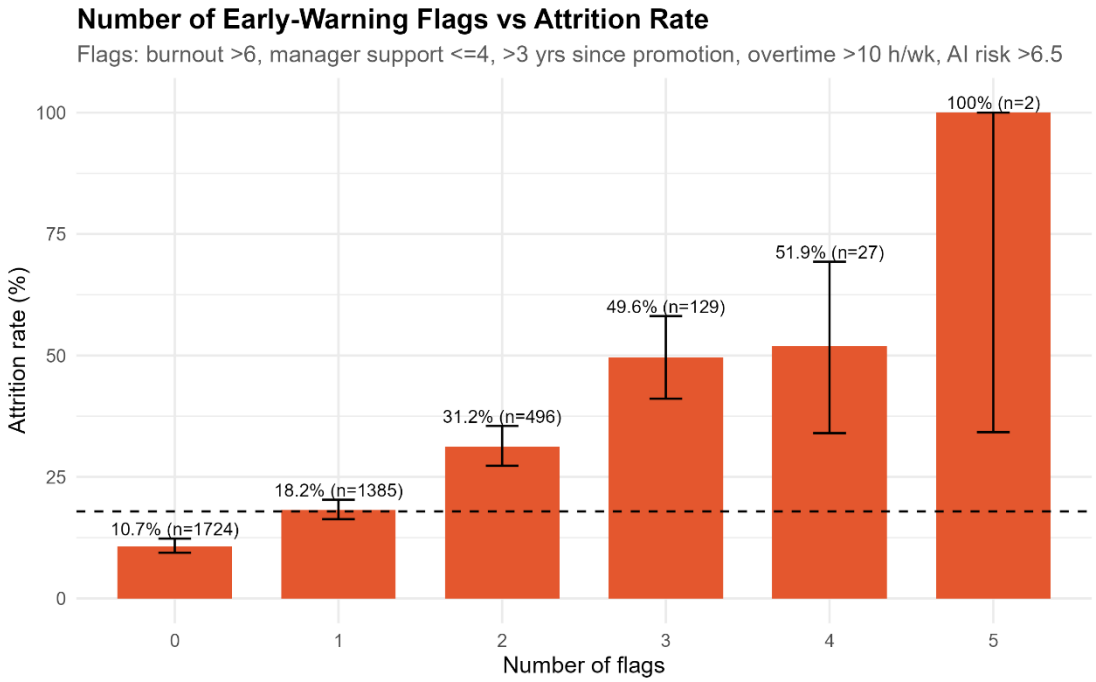
Burnout's effect on attrition is not linear. Attrition sits below 20% up till burnout score of 4 before climbing steeply to 70%. Employee headcount distribution also shows the issue is concentrated rather than workforce-wide, with a minority having crossed the threshold of strain where leaving becomes likely, so return on addressing these issues is highest before the threshold is crossed.

Figure 3: Attrition probability rises with burnout



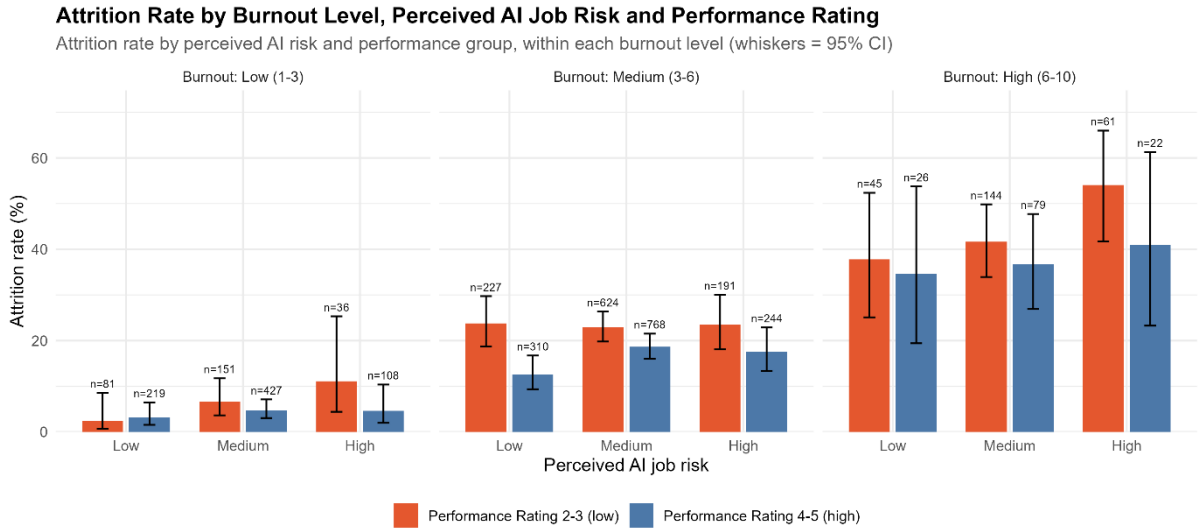
Rather than investing for all 3,763 staff, targeting the roughly 17.5% with two or more warning flags is more efficient as they account for 35% of all the leavers.

Figure 4: Attrition rises with the number of early warning flags



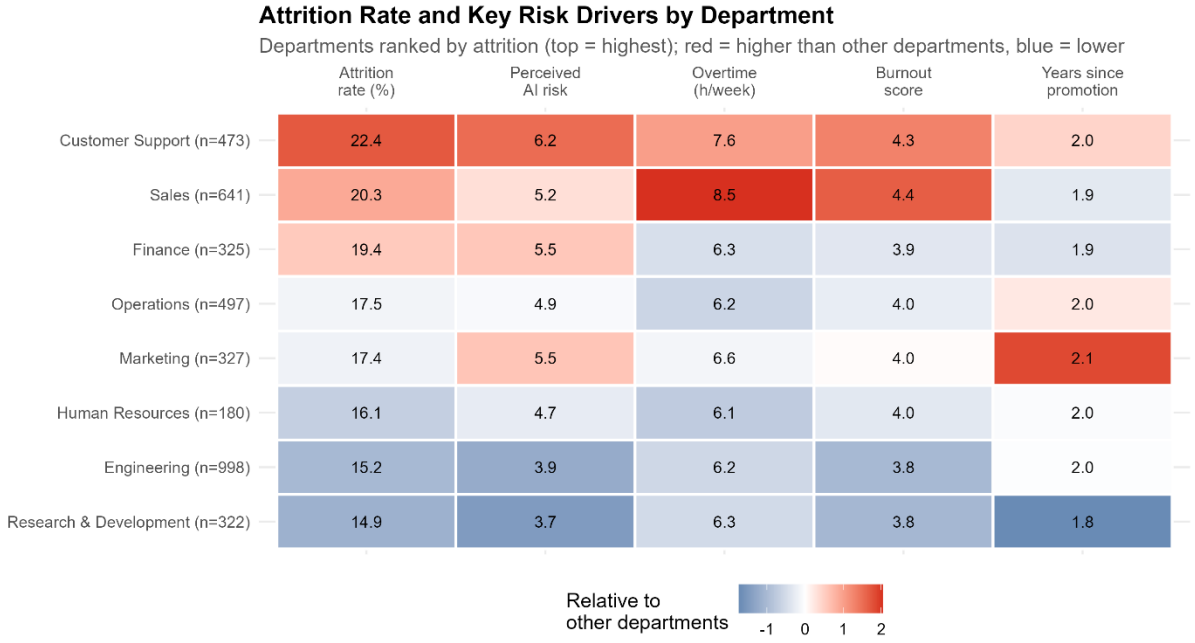
Anxiety from AI job risk does matter however rather than a direct cause, it acts as a multiplier. Attrition is barely affected by it yet high-burnout staff show a difference of roughly 55% for high AI risk vs. 37% for the lowest risk. This indicates that fear of being automated out of a job acts as an accelerant, adding onto existing strain, rather than being a standalone driver.

Figure 5: Burnout sets level of attrition; performance and AI risk shift it modestly



Attrition also differs by department. Customer Support carries the heaviest risk profile with the highest AI-risk score of 6.2, while Sales has the highest overtime driven of 8.5 hours per week. That means a one-size-fits-all retention plan would likely fail to address each department's primary risk outside of burnout.

Figure 6: Customer Support and Sales carry the heaviest risk profile



4.1 Dataset Suitability

The cleaned dataset is suitable for descriptive and predictive work with ~3,700 rows and ~670 leavers however unsuitable for causal inference as it has no data on leave dates, so analysis can only show associations rather than causes. There are also three limitations that constrain interpretation: no employee scored a 1 on performance, non-binary is in the data dictionary but never in the data, and there is heavy overlap between burnout, work-life balance and engagement, making it hard to isolate individual effects.

5. Recommendations

The company should focus investments into strengthening management and workload control. Manager training and lowering workload targets addresses the primary drivers that actually impact attrition before employees pass the threshold where leaving becomes plausible. Testing this in the highest-risk Customer Support and Sales departments would be a good low-cost option before a full rollout.

The company should also utilize existing data to test and deploy a tiered early warning system, where staff with two or more warning flags provide managers with a clear list of who to check in with. What they do next should depend on the department: reduce workload in Sales, provide support around AI concerns in Customer Support. Since the flag thresholds were based off the same sample, they should be tested further on new data before real use.